7 This question is about the magnetic fields around bar magnets. Fig. 7.1 shows two positions used by a student doing an experiment.



Fig. 7.1

(a) Fig. 7.2 shows a magnet, labelled magnet 1, placed on position 1.



Fig. 7.2

On Fig. 7.2, draw lines to show the pattern of the magnetic field produced by magnet 1. Place arrows on the lines to show the direction of the field. [3]

(b) Magnet 1 is removed from position 1. Fig. 7.3 shows another magnet, labelled magnet 2, placed on position 2.



Fig. 7.3

On Fig. 7.3, draw, at the right-hand end of position 1, a line with an arrow to show the direction of the magnetic field produced by magnet 2. [1]

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(c) Fig. 7.4 shows magnet 1 placed on position 1 and magnet 2 placed on position 2.



Fig. 7.4

(i) State the direction of the force that the N pole of magnet 2 exerts on the N pole of magnet 1.

..... [1]

(ii) Justify your answer to **(c)(i)**.

.....

..... [1]

[Total: 6]